

Roles and Flows

The three roles, the order lifecycle, and the end-to-end journeys — written down so both developers build against the same understanding. This is the *behaviour* the modules have to produce.

1. The three roles

Roles live on the user record and are enforced on the server. The demo's role-switcher menu does not survive into this build.

Customer

Who	The public. Self-registered, personal or company.
Sees	Their own orders only — booking, order list, order detail, tracking, notifications, profile
Can do	Sign up, book, pay online, view their own orders, track after dispatch
Cannot	See anyone else's orders · reach any ops or agent screen · trigger any lifecycle action · see internal statuses, staff names, or cost breakdowns
Never sees	The words <code>weighed</code> , <code>settled</code> , <code>ready_for_docket</code> . Those are internal.

Pickup Agent

Who	Bombino field staff, on a phone, in the street
Sees	Pickups nobody has claimed · their own claimed jobs · the detail needed to do the job (address, contact, parcel, whether money is owed)
Can do	Claim an unclaimed pickup · mark out for pickup · collect payment at the door if the customer chose pay-at-pickup · mark picked up · mark received at hub
Cannot	Weigh · settle · reprice · generate a shipment · see orders that aren't theirs or aren't available · touch an order after it reaches the hub
Key rule	Agents self-assign . There is no dispatcher and no assignment screen anywhere in this build.

Operations / Admin

Who	Bombino office and hub staff
Sees	Every live order, on the board, grouped by stage
Can do	Mark drop-offs received · weigh · reprice · collect payment at the hub · settle · generate the shipment and tracking number
Cannot	Claim pickups on an agent's behalf · amend a shipment after it's registered with ITD (nobody can — ITD forbids it)
Key rule	Ops is the system of record for every order between booking and dispatch.

Super Admin (*only if confirmed in scope*)

Manages internal users — creates agent and ops accounts, assigns roles, deactivates leavers. Everything Ops can do, plus user management.

2. The order lifecycle

Every status, who moves it, and what the customer sees.

#	Status	Moved by	What just happened	Customer sees
1a	pickup_requested	<i>(system, at booking)</i>	Customer booked and asked for collection	Pickup requested
1b	awaiting_dropoff	<i>(system, at booking)</i>	Customer booked and will drop off themselves	Awaiting drop-off
2	agent_accepted	Agent	An agent claimed the job	Pickup confirmed
3	out_for_pickup	Agent	Agent is on the way	Agent on the way
4	picked_up	Agent	Parcel collected from the customer	Parcel picked up
5	received_at_hub	Agent (pickup path) or Ops (drop-off path)	Parcel is physically at the hub	Arrived at Bombino hub
6	weighed	Ops	Real weight recorded, price recalculated	<i>(nothing — internal)</i>
7	settled	Ops	Money difference resolved	<i>(nothing — internal)</i>
8	ready_for_docket	Ops	All checks passed, queued for registration	<i>(nothing — internal)</i>
9	dispatched	Ops	Registered with ITD, real tracking number issued	In transit — now trackable
—	cancelled	Ops or Customer	Order stopped before it reached the hub	Cancelled

Three statuses are deliberately silent. Between arriving at the hub and being dispatched, the customer sees no change. From their point of view the parcel is at the hub and nothing has happened yet. Firing notifications for weighing and settling would be noise.

The hard gate: an order cannot reach `ready_for_docket` unless the weight has been captured **and** the payment is satisfied. Pay-on-delivery is the single exception — it passes the payment check by design.

3. End-to-end journeys

Flow A — Personal customer, pickup, pays online

The most common path.

Step	Actor	What happens
1	Customer	Opens the app, taps Create Account, chooses Personal
2	Customer	Enters phone number, receives a real SMS code, enters it
3	Customer	Fills in name, email, address, and Aadhaar details (format-checked, not verified)
4	System	Creates the Bombino account, links it to the shared retail ITD account, logs them in
5	Customer	Books: origin, destination, parcel details, estimated weight
6	Customer	Chooses Pickup , picks a date and a 3-hour window
7	Customer	Chooses Pay now , completes payment through the gateway
8	System	Creates a Bombino order, shows an Order ID . <i>No tracking number exists yet.</i>
9	Agent	Sees the job in the available list, claims it
10	Agent	Marks out for pickup, arrives, collects the parcel, marks picked up
11	Agent	Drops it at the hub, marks received at hub. The job leaves the agent's list.
12	Ops	Sees it on the board under <i>At Branch</i> , weighs it — it's 3 kg, not the 2 kg booked
13	System	Reprices and shows collect ₹340 more
14	Ops	Collects the difference, marks it settled
15	Ops	Generates the shipment — ITD returns a real tracking number
16	Customer	Sees In transit , and the order becomes trackable

Flow B — Company customer, drop-off, pays at the hub

Step	Actor	What happens
1	Customer	Creates an account, chooses Company
2	Customer	Verifies phone by SMS, enters company details and GST (format-checked only)
3	System	Creates a real ITD customer via <code>add_customer</code> , saves the identity, logs them in
4	Customer	Books, chooses Drop-off , chooses Pay at drop-off
5	System	Creates the order with an Order ID. Status: <i>awaiting drop-off</i>
6	Customer	Brings the parcel to the hub
7	Ops	Marks it received at hub
8	Ops	Weighs it, price is recalculated
9	Ops	Takes payment at the counter, marks it collected
10	Ops	Settles, then generates the shipment
11	Customer	Order becomes trackable

No agent is involved anywhere in this flow.

Flow C — Pay on delivery (COD)

Identical to Flow A or B up to settling, with one difference that matters:

Step	What happens
At booking	Customer chooses Pay at delivery . The order is flagged COD. No payment is taken.
At weigh	The price is still recalculated, and the difference is still shown
At settle	The order settles even though nothing has been collected. COD passes the payment check by design
At docket	The shipment generates normally
Afterwards	The order stays flagged COD for the accounts team

COD must never block a shipment. If a COD order gets stuck at settle, that's a bug.

Flow D — Weight came in lower

Step	What happens
Booked at 3 kg, paid in advance; actual weight 2 kg	
At weigh	System shows refund due ₹180
At settle	Ops records that a refund is owed
This phase	The refund is flagged, not processed — accounts handle it outside the system
Then	The order settles and generates normally

4. When money is collected

Method	Collected when	Collected by	Built by
Pay now	At booking	Payment gateway	Aditya
Pay at pickup	At the customer's door	Pickup agent	Aditya (agent app)
Pay at drop-off	At the hub counter	Ops	Arbaaz (ops console)
Pay at delivery (COD)	Never, by us	Nobody — flag only	Aditya

Three rules that apply across all of them:

1. **The amount is an estimate until the parcel is weighed.** Even a fully prepaid order can end up owing more, or being owed a refund. Everything reconciles at settle.
2. **Payment is entirely Bombino-side.** No payment information is ever sent to ITD.
3. **COD is the only method that doesn't gate the shipment.**

5. Handoff points

Three moments where an order changes hands. Each one needs to be clean and one-directional — these are where bugs will live.

Handoff	From → To	At	What must be true
Booking to fulfilment	Customer → Agent (<i>or Ops</i>)	Order created	The order carries everything downstream needs: consignee, pickup address, amount, pickup choice and slot, payment method, booked weight. Ops has no other source for any of it.
Field to hub	Agent → Ops	<code>received_at_hub</code>	The job disappears from the agent's list and appears on the ops board. The agent can no longer act on it.
Internal to live	Ops → ITD	<code>dispatched</code>	The shipment is registered with final values and becomes immutable. Everything before this is editable; nothing after it is.

Where two people can collide

Situation	Required behaviour
Two agents claim the same pickup at the same moment	Exactly one wins. The other gets a clear message, not a silent failure or a double-assigned parcel.
Ops advances an order while the customer is looking at it	The customer's view updates without them doing anything, and a refresh shows current truth.
Someone tries to generate the shipment twice	Impossible, not merely unlikely. A duplicate shipment cannot be undone in ITD.
Two ops staff open the same order	Both see the same state; the second action to land reflects the first.